Student ID:	24301500	**Lab Section:**	7B
Name:	Magidul Islam Kabbo	**Lab Group:**	6

Experiment-02

Study of Op-Amp: Introduction to Op-Amp, Comparator Circuits, Non-Inverting Amplifier, Inverting Amplifier, Inverting Summing Amplifier & their VTCs

BRAC UNIVERSITY
Inspiring Excellence

CSE251 - Electronic Devices and Circuits Lab

Objective

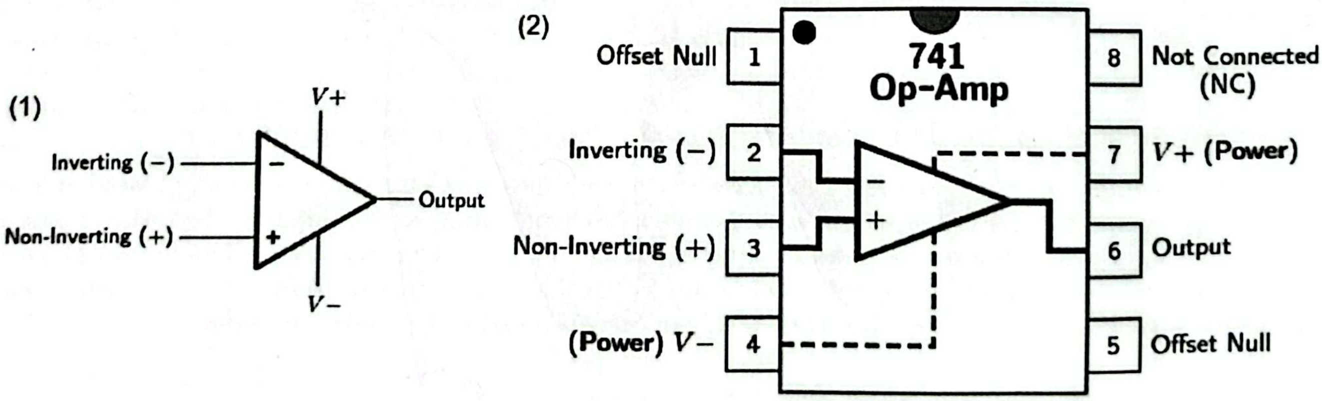
1. To understand the basic principles and characteristics of an Operational Amplifier (Op-Amp)
2. To understand Operational Amplifier (Op-Amp) as a Comparator and investigate its use
3. To investigate the use of Operational Amplifier (Op-Amp) as Non-Inverting Amplifier, Inverting Amplifier and Inverting Summing Amplifier

Equipment

1. Op-Amp (uA741)
2. Resistance ($1k\Omega$, $2.7k\Omega$, $10k\Omega$)
3. DC Power Supply
4. Function Generator
5. Digital Multimeter
6. Trainer Board, Breadboard, Chords and Wires

Background Theory

Introduction



(1) Op-Amp Simplified Circuit Symbol (2) Op-Amp IC Pin Diagram

One of the most widely used electronic devices in linear applications is the Operational Amplifier, commonly known as the Op-Amp. An Op-Amp is an integrated circuit that amplifies the difference between two input voltages and produces a single output. We can also do various mathematical operations like addition, subtraction, multiplication, integration, differentiation etc. with the help of Op-Amp. With the addition of suitable external components, Op-Amp can be used for a variety of applications. The figure above shows the simplified circuit symbol of an Op-Amp. There are 2 terminals for input, 1 terminal for output and 2 t

Task-05: Report

1. Attach the signed Data Sheet (if any)
2. Attach the captured images (if any)
3. Answer the questions in the "Test Your Understanding" section
4. Add a brief Discussion regarding the experiment. For the Discussion part of the lab report, you should include the answers of the following questions in your own words:
   - What did you learn from this experiment? — How to connect and use opamp in three ~~two ways~~ ways
   - What challenges did you face and how did you overcome the challenges? (if any)
   - What mistakes did you make and how did you correct the mistakes? (if any) — wrong connections, changed connection to fix it
   - How will this experiment help you in future experiments of this course?
     helped me get a foundational idea on an operational circuit element.

Data Sheet

For Task-02:

Quantity	Value
Value of R_1 using multimeter	0.981
Value of R_2 using multimeter	2.65
Input Amplitude from Oscilloscope, $v_I = \frac{Pk-Pk}{2}$ (use the Measure button of the Oscilloscope)	~~2.08~~ 1.04
Output Amplitude, $v_O = \frac{Pk-Pk}{2}$ (use the Measure button of the Oscilloscope)	~~7.92~~ 3.92
Output Amplitude, $v_O = (1 + \frac{R_2}{R_1}) \times v_I$ (theoretical calculation)	3.7

For Task-03:

Quantity	Value
Value of R_1 using multimeter	.961
Value of R_2 using multimeter	2.65
Input Amplitude from Oscilloscope, $v_I = \frac{Pk-Pk}{2}$ (use the Measure button of the Oscilloscope)	1
Output Amplitude, $v_O = \frac{Pk-Pk}{2}$ (use the Measure button of the Oscilloscope)	2.76
Output Amplitude, $v_O = -(\frac{R_2}{R_1}) \times v_I$ (theoretical calculation)	-2.7

For Task-04:

Quantity	Value
Value of R_1 using multimeter	9.88
Value of R_2 using multimeter	9.9
Value of R_F using multimeter	9.96
from multimeter, v_{I1}	0.998
from multimeter, v_{I2}	1.971
Output Amplitude from multimeter, v_O	-2.995
Output Amplitude from equation, $v_O = -(\frac{R_F}{R_1} \times v_{I1} + \frac{R_F}{R_2} \times v_{I2})$	-2.989

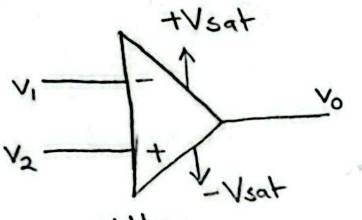
22/10/25

Signature of the lab faculty

Test Your Understanding

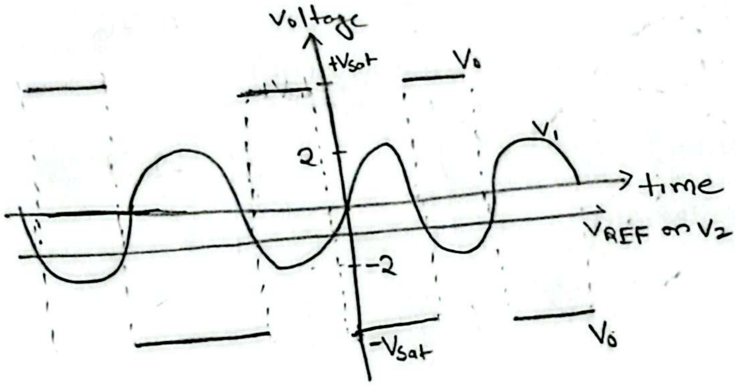
Answer the following questions:

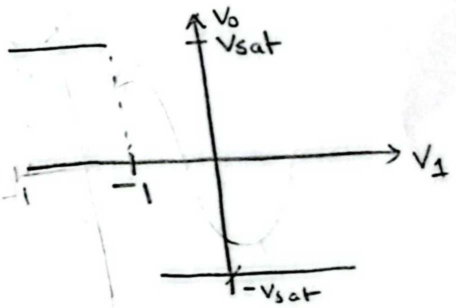
1. You are given an Op-Amp comparator with v_1 = 4 V (p-p) sine wave and $v_2 = V_{REF}$ = -1 V. Draw the waveform of v_1, v_2 and v_O in the same graph with proper labels.
Answer:



when $V_2 > V_1$
$V_o = +V_{sat}$

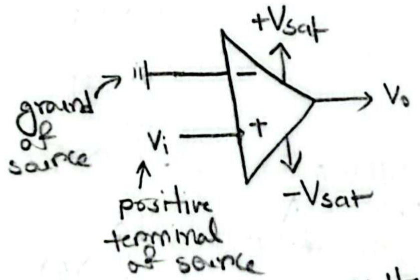
when $V_2 < V_1$
$V_o = -V_{sat}$

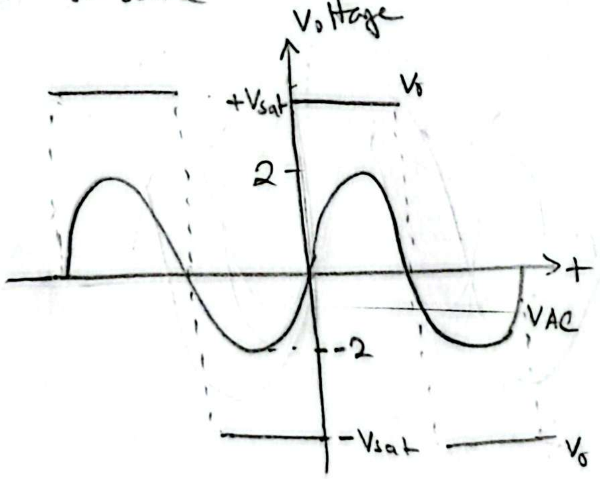


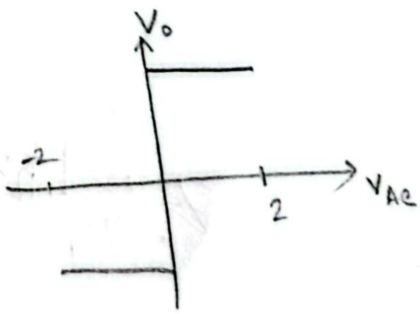


2. You are given an Op-Amp comparator with v_i = 2 sin(t), where the two ends of the input voltage, v_i (sinusoidal source) is connected to v_1 and v_2 of Op-Amp. Draw the waveform of v_{AC} and v_O in the same graph with proper labels.
Answer:

$V_i = 2\sin t$

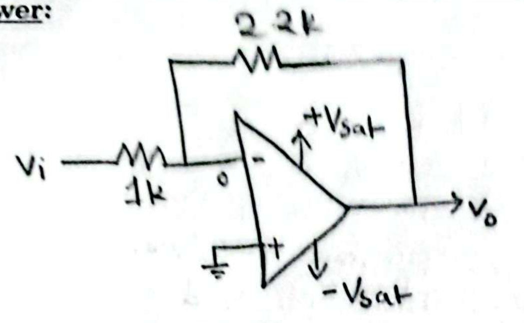




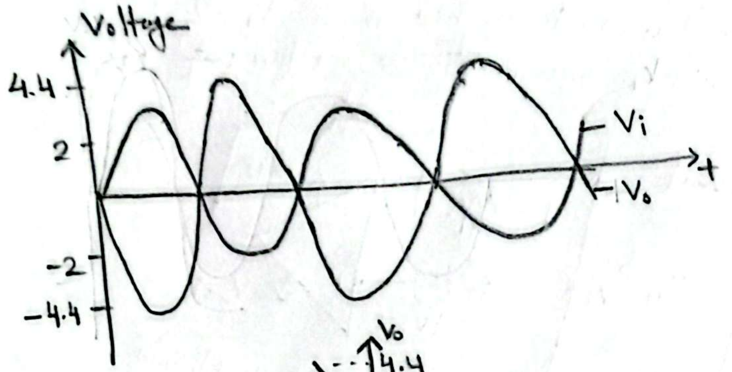


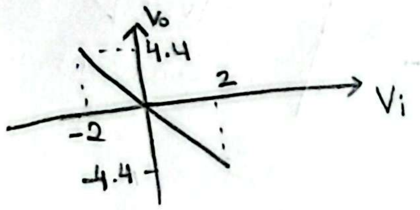
3. You are given an inverting amplifier with $v_I = 4$ V (p-p) sine wave, $R_1 = 1\ k\Omega$, $R_2 = 2.2\ k\Omega$. Draw the waveform of v_I and v_O in the same graph with proper labels.

Answer:



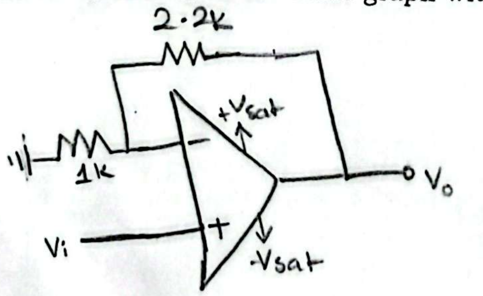
$$V_o = -\frac{R_F}{R_1} \times V_i = -2.2V_i$$



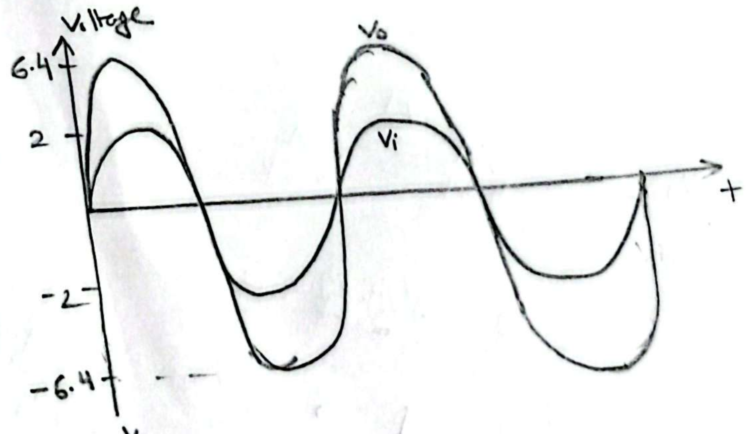


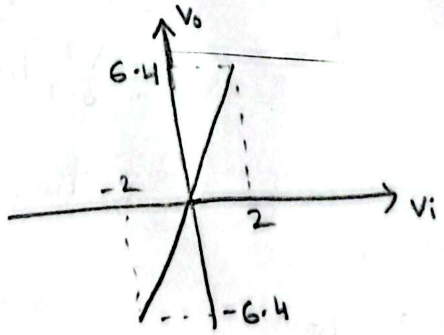
4. You are given a non-inverting amplifier with $v_I = 4$ V (p-p) sine wave, $R_1 = 1\ k\Omega$, $R_2 = 2.2\ k\Omega$. Draw the waveform of v_I and v_O in the same graph with proper labels.

Answer:



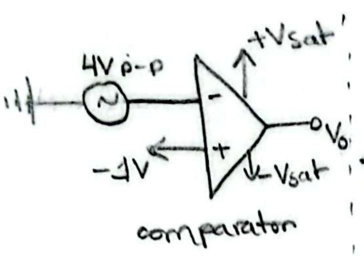
$$V_o = \left(1 + \frac{R_F}{R_1}\right) V_i$$
$$= 3.2V_i$$

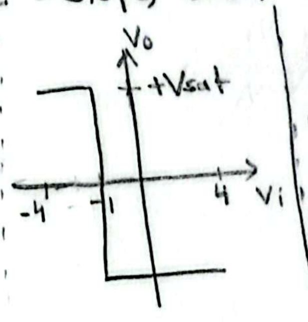


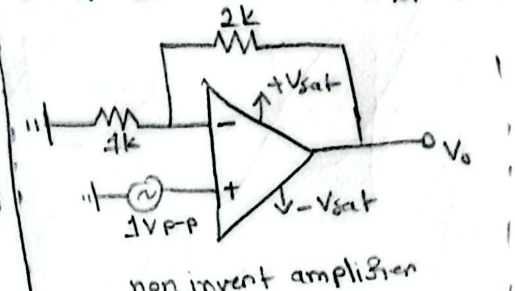


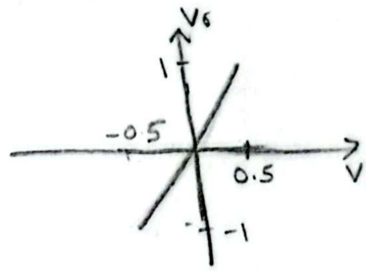
5. Discuss the differences in behavior between the comparator's switching operation and the amplifier's linear scaling. For the comparator, use a 4 V (peak-to-peak) sine wave to the inverting terminal with a fixed reference of -1 V to the non inverting terminal. For the non-inverting amplifier, set the gain to 2 with a 1 V (peak-to-peak) sine wave input. For each circuit, draw the VTC (i.e., v_{out} vs. v_{in}) on separate graphs with proper labels and discuss accordingly.

Answer: In comparator, the output V_o is $+V_{sat}$ when V_d in input is positive, else its $-V_{sat}$. An Op Amp can amplify a raw V_d to upwards of 10^6 or ∞ in theory, since this value is so high, for very little V_d, this V_o reaches its limit which is $+V_{sat}$ or $-V_{sat}$. So, in comparator V_o almost instantly reaches $+V_{sat}$ or $-V_{sat}$ but in amplifiers, resistors are used to reduce amplification so, the V_d required is much higher to reach $+V_{sat}$, so in VTC we can see a slope, which corresponds to either $(\frac{-R_F}{R_1}+1)$ or $-\frac{R_F}{R_1}$.



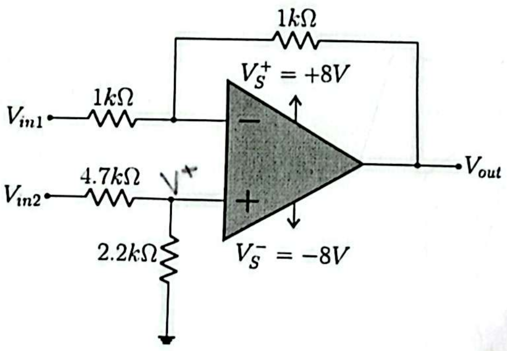






6. Deduce the value of V_{out} from the circuit below.

[Hints: Consider the current towards the inverting and non-inverting terminals of Op-Amp is zero, and the voltage of the inverting terminal equals the voltage of the non-inverting terminal]



Answer:

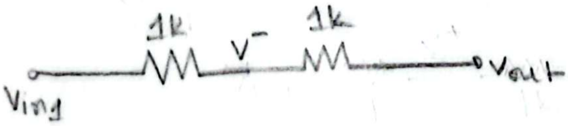
$$V_{in_2} - 0 = (I \times 4.7) + 2.2I$$

$$\frac{V_{in_2}}{6.9k} = I \Rightarrow \quad V_{in_2} - V^+ = 4.7I = \frac{4.7 \times V_{in_2}}{6.9}$$

$$V_{in_2} - \frac{4.7}{6.9}V_{in_2} = V^+$$

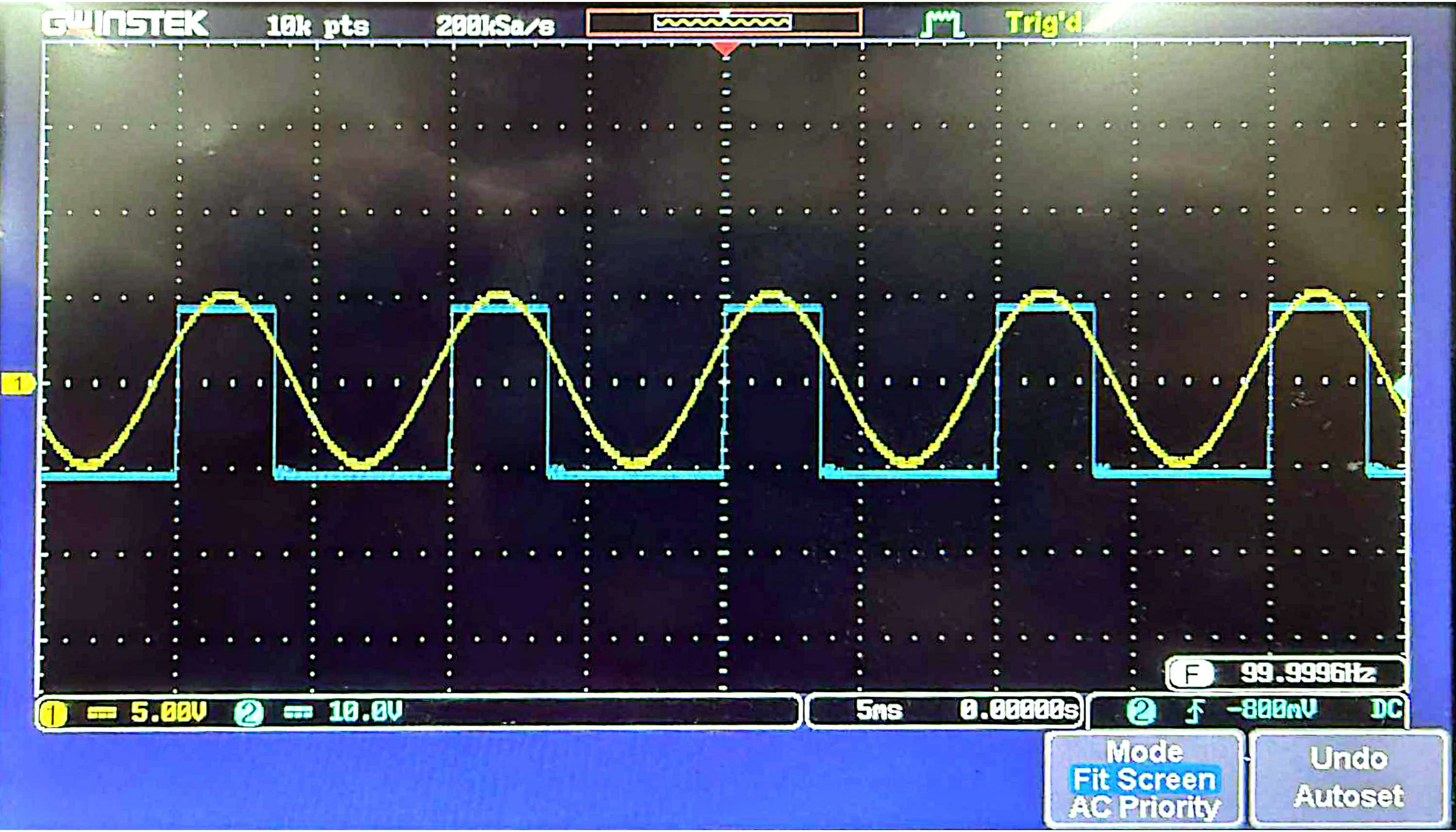
$$\frac{2.2}{6.9}V_{in_2} = V^+$$

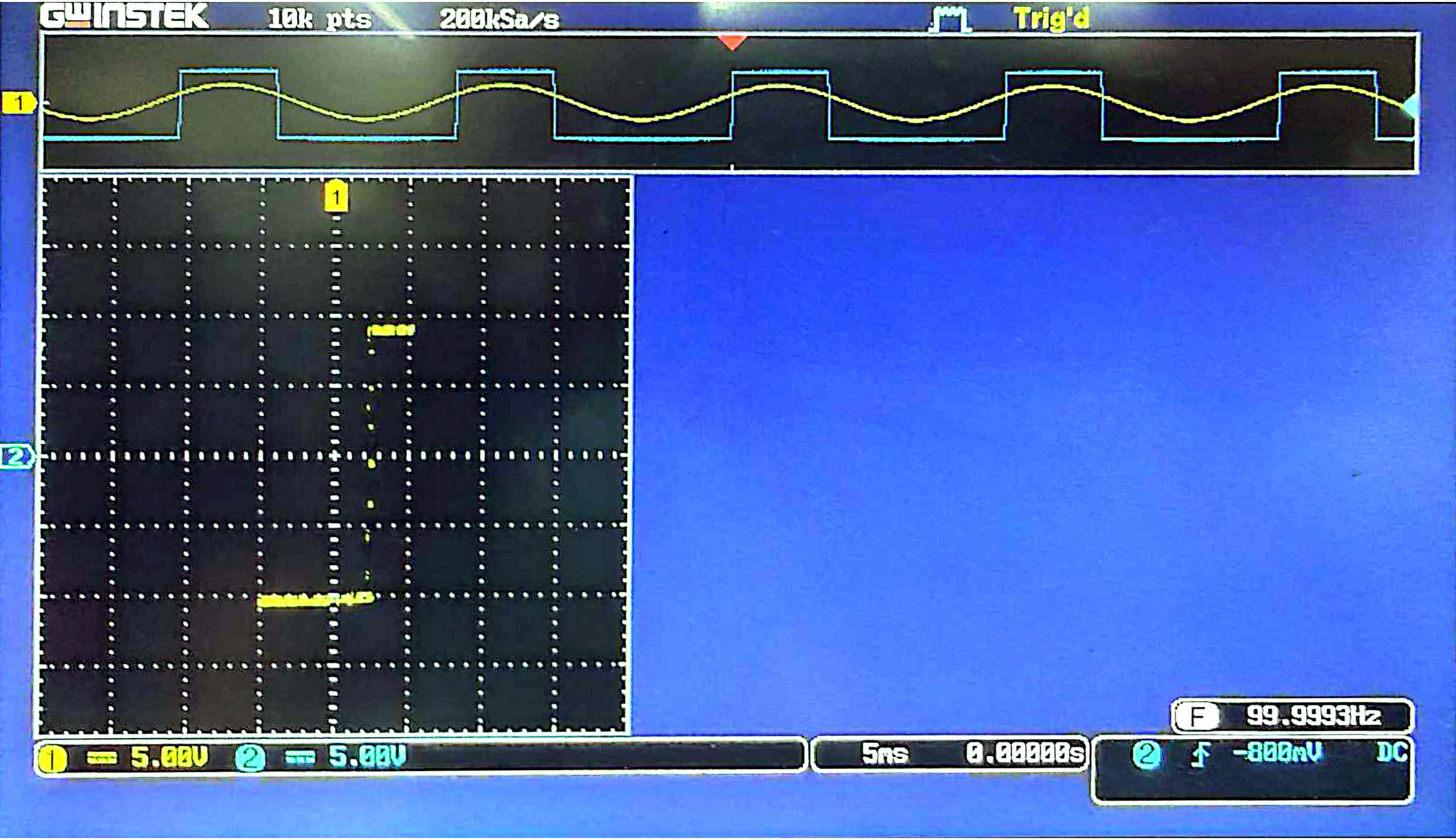
$$V^+ = V^- = \frac{2.2}{6.9}V_{in_2}$$



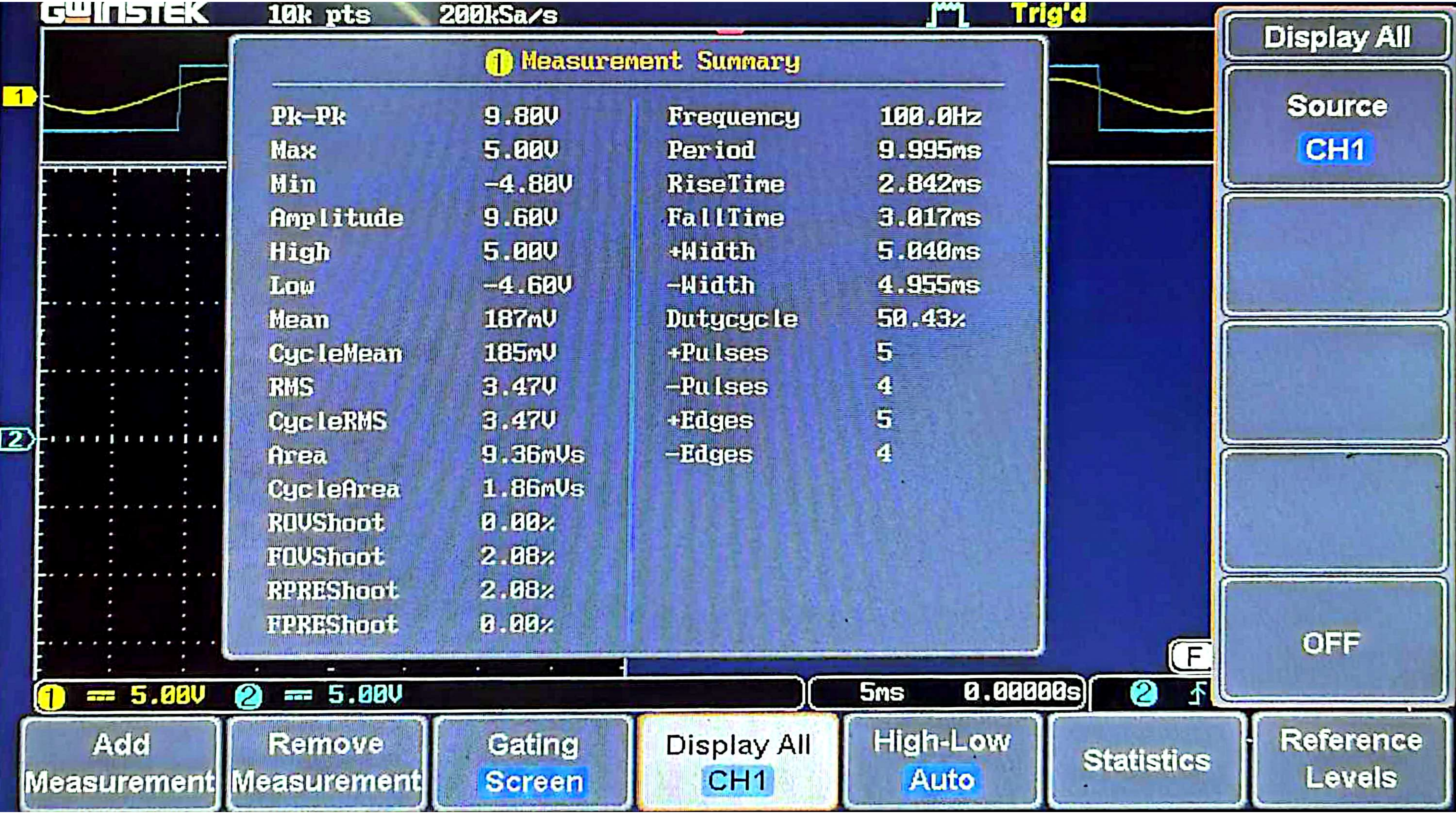
$$\frac{V_{in_1} - V^-}{1} = \frac{V^- - V_{out}}{1} \Rightarrow 2V^- - V_{out} = V_{in_1}$$

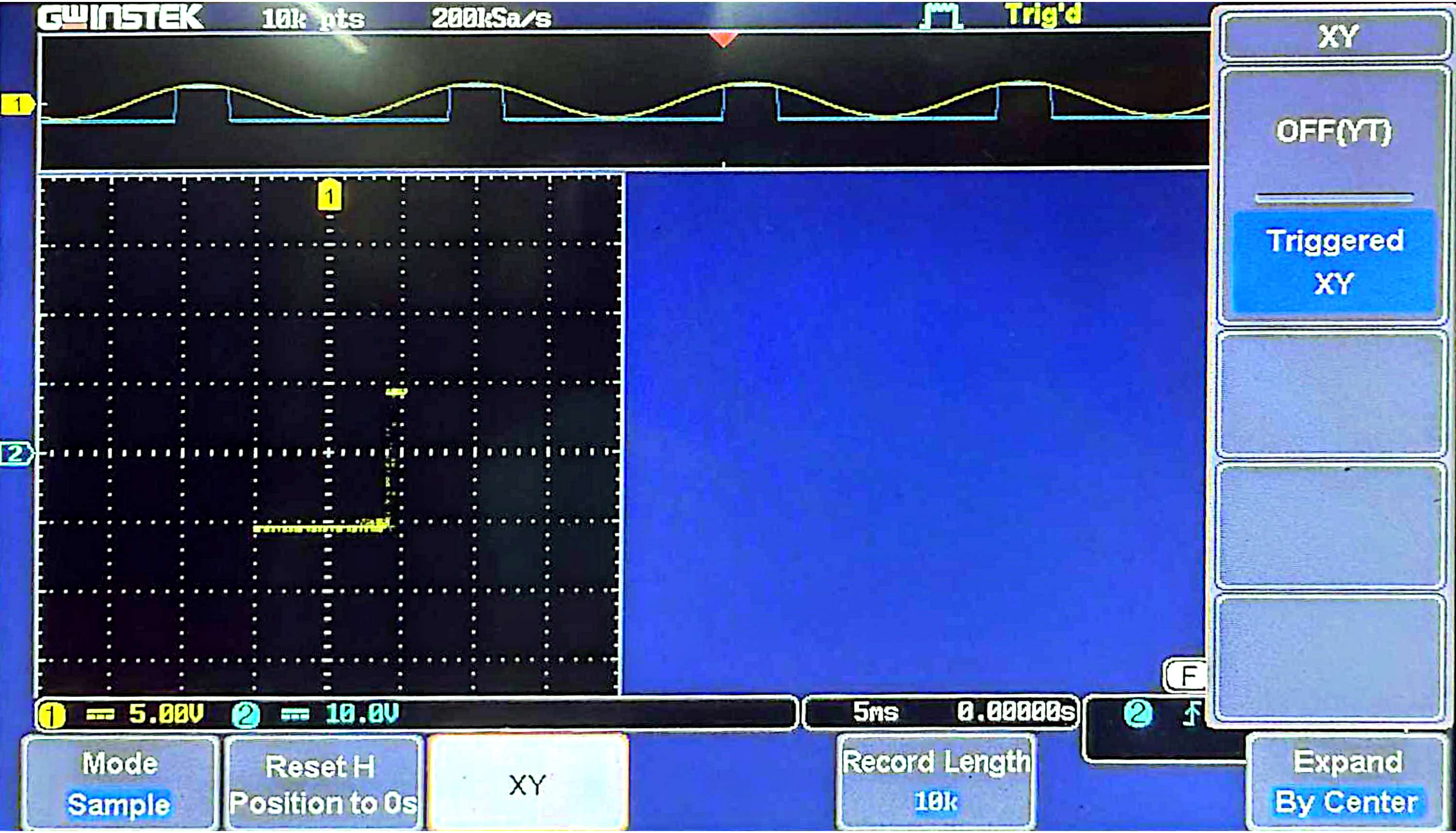
$$2V^- - V_{in_1} = V_{out} \Rightarrow V_{out} = \frac{4.4}{6.9}V_{in_2} - V_{in_1} \ \boxed{\text{ANS}}$$

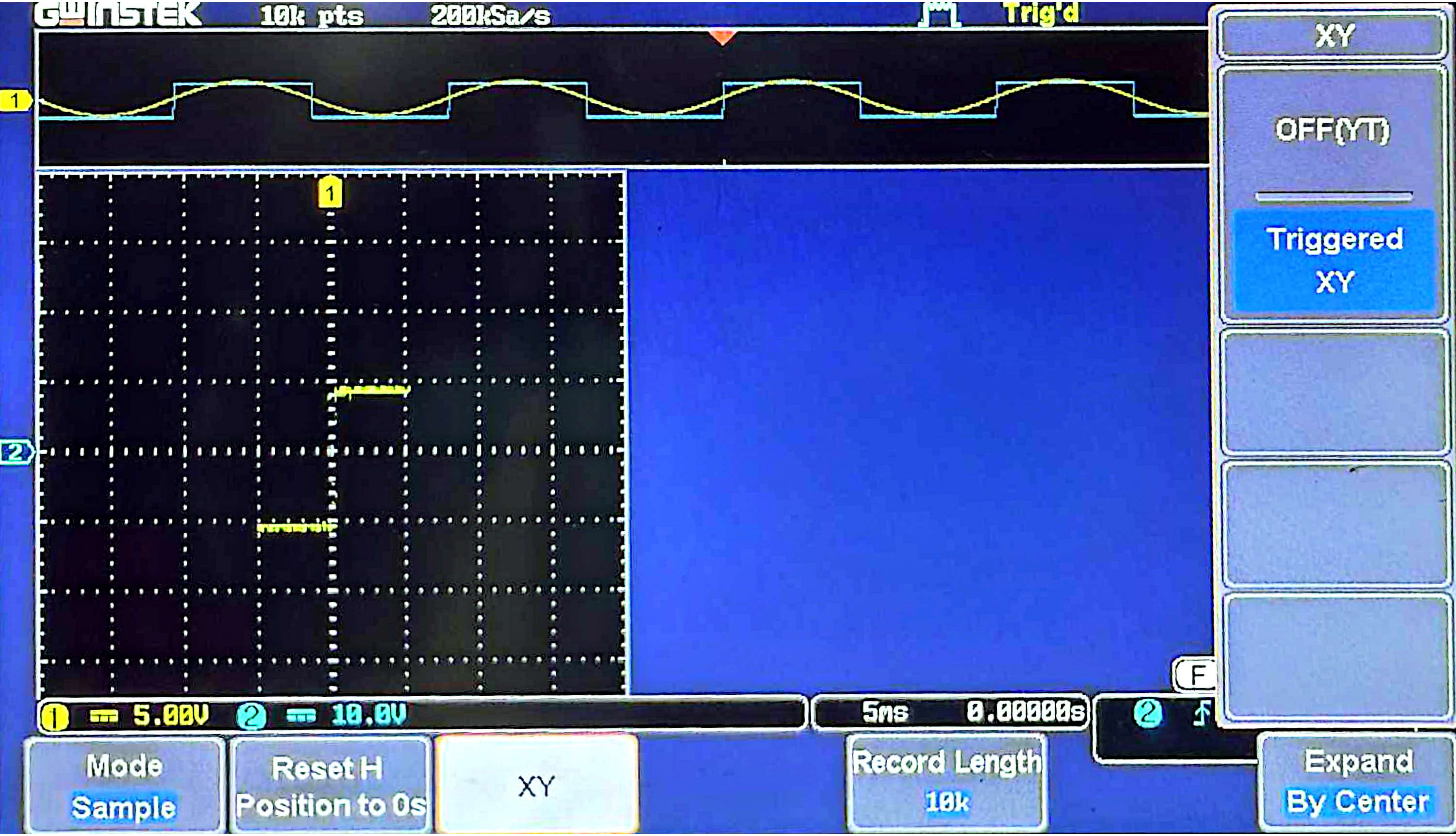

GWINSTEK
10k pts
200kSa/s
Trig'd
99.9996Hz
1 5.00V
2 10.0V
5ms
0.00000s
2 -800mV
DC
Mode
Fit Screen
AC Priority
Undo
Autoset



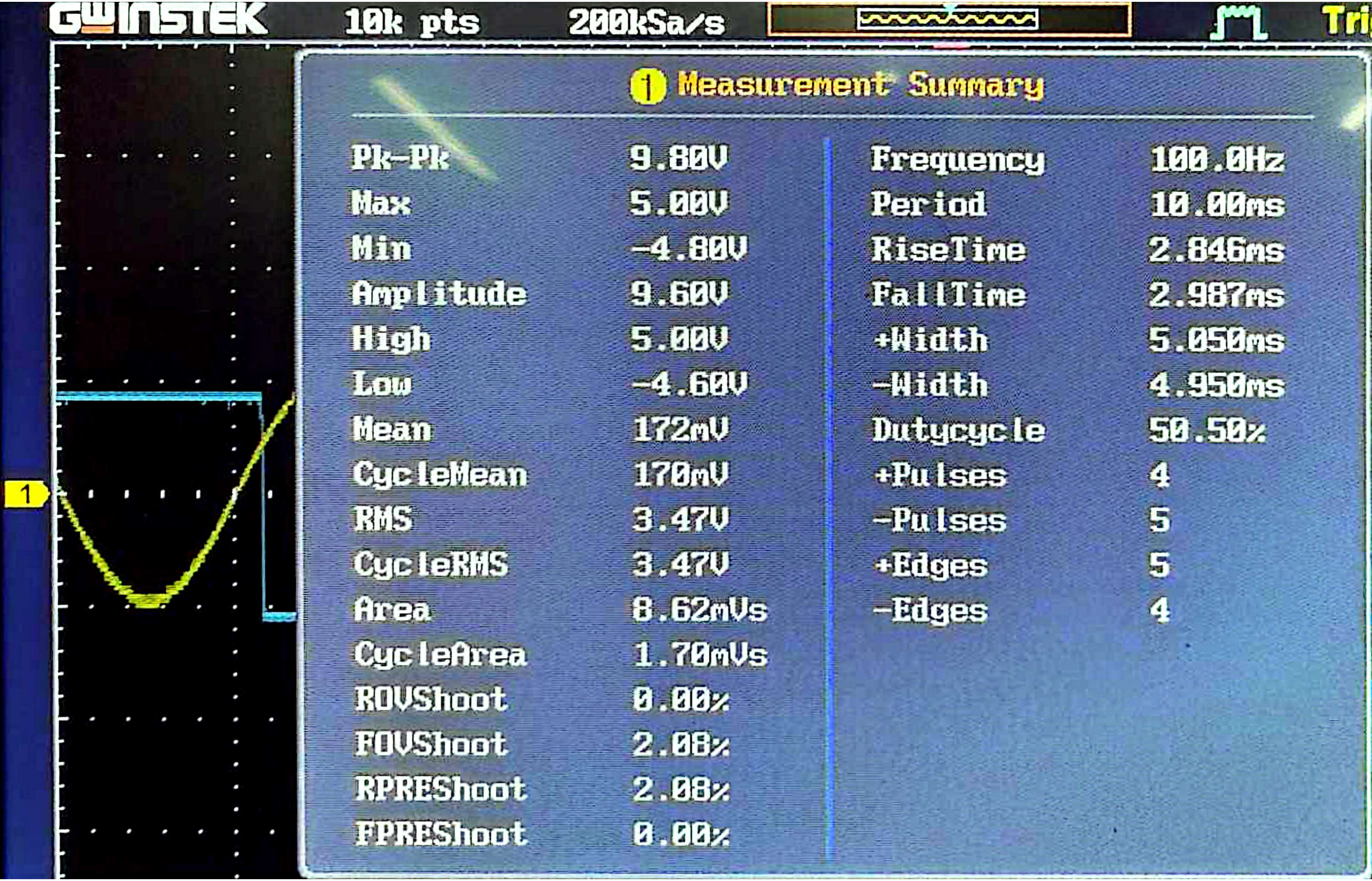
GWINSTEK
10k pts
200kSa/s
Trig'd
1
1
2
99.9993Hz
1 5.00V
2 5.00V
5ms
0.00000s
2 -800mV
DC

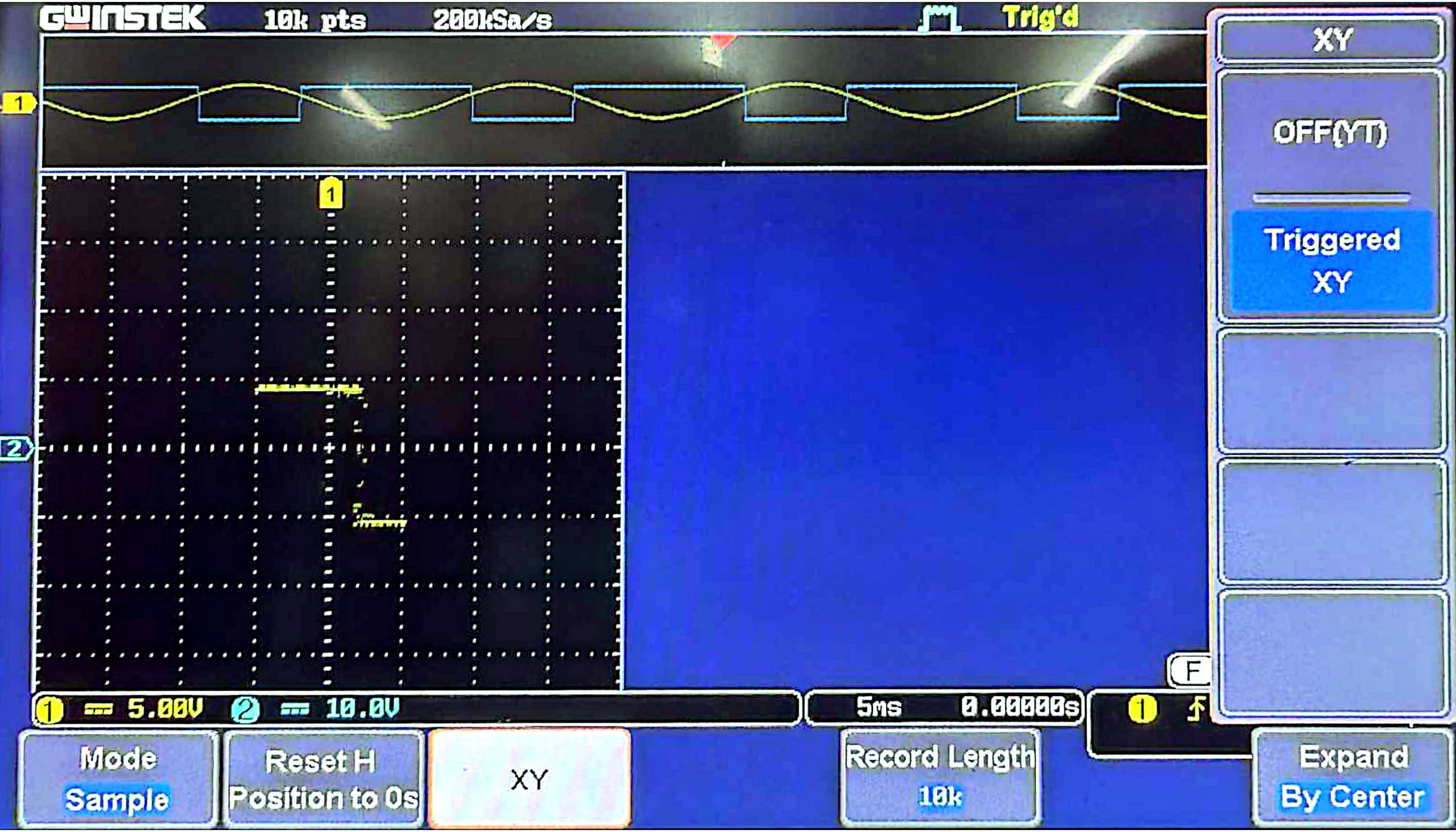

GW INSTEK
10k pts
200kSa/s
Trig'd
Measurement Summary
Pk-Pk 9.80V
Max 5.00V
Min -4.80V
Amplitude 9.60V
High 5.00V
Low -4.60V
Mean 187mV
CycleMean 185mV
RMS 3.47V
CycleRMS 3.47V
Area 9.36mVs
CycleArea 1.86mVs
ROVShoot 0.00%
FOVShoot 2.08%
RPREShoot 2.08%
FPREShoot 0.00%
Frequency 100.0Hz
Period 9.995ms
RiseTime 2.842ms
FallTime 3.017ms
+Width 5.040ms
-Width 4.955ms
Dutycycle 50.43%
+Pulses 5
-Pulses 4
+Edges 5
-Edges 4
Display All
Source CH1
OFF
1 5.00V 2 5.00V
5ms 0.00000s
Add Measurement
Remove Measurement
Gating Screen
Display All CH1
High-Low Auto
Statistics
Reference Levels

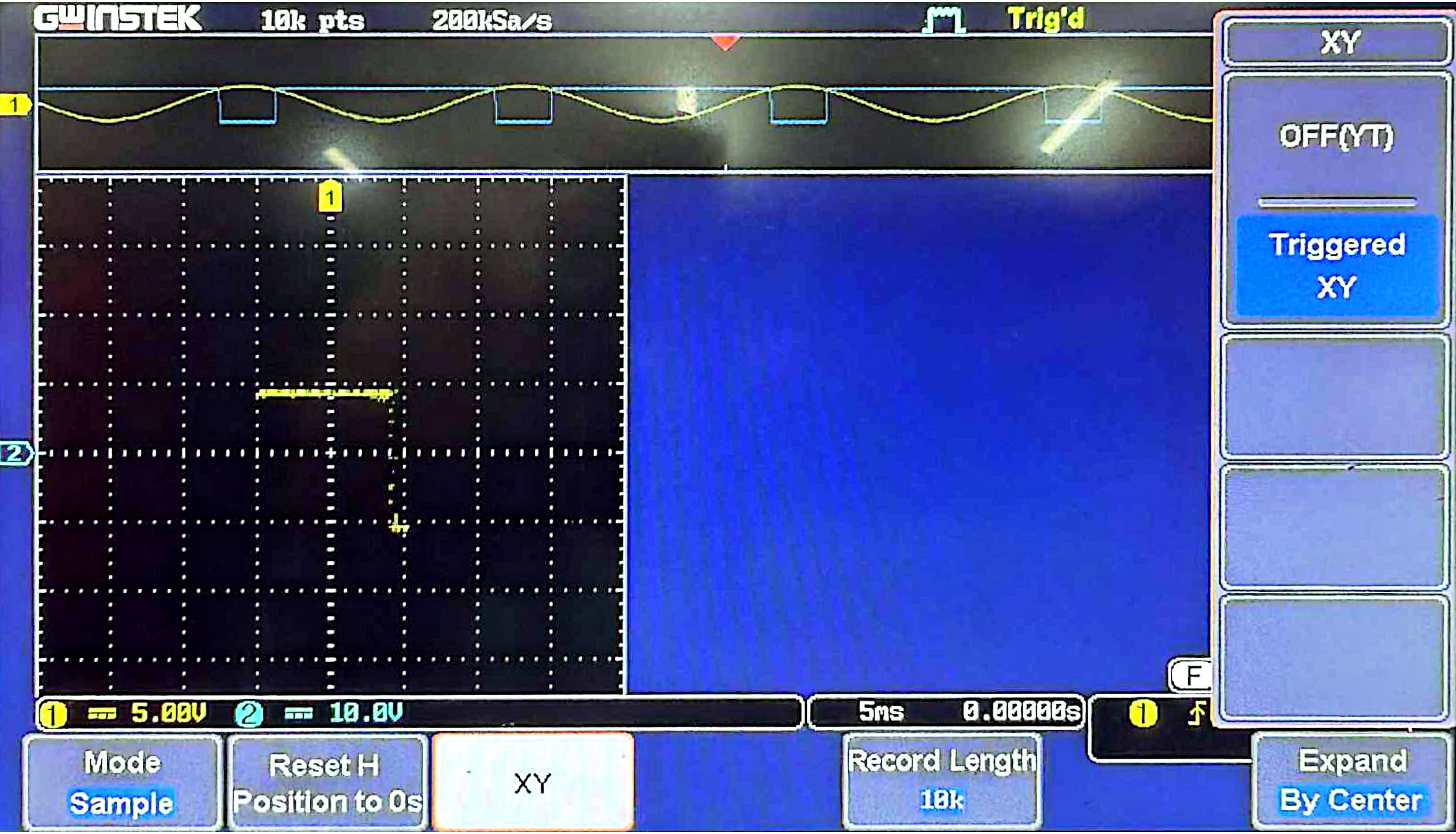

GWINSTEK
10k pts
200kSa/s
Trig'd
XY
OFF(YT)
Triggered XY
1
2
1 5.00V
2 10.0V
5ms
0.00000s
F
Mode Sample
Reset H Position to 0s
XY
Record Length 10k
Expand By Center

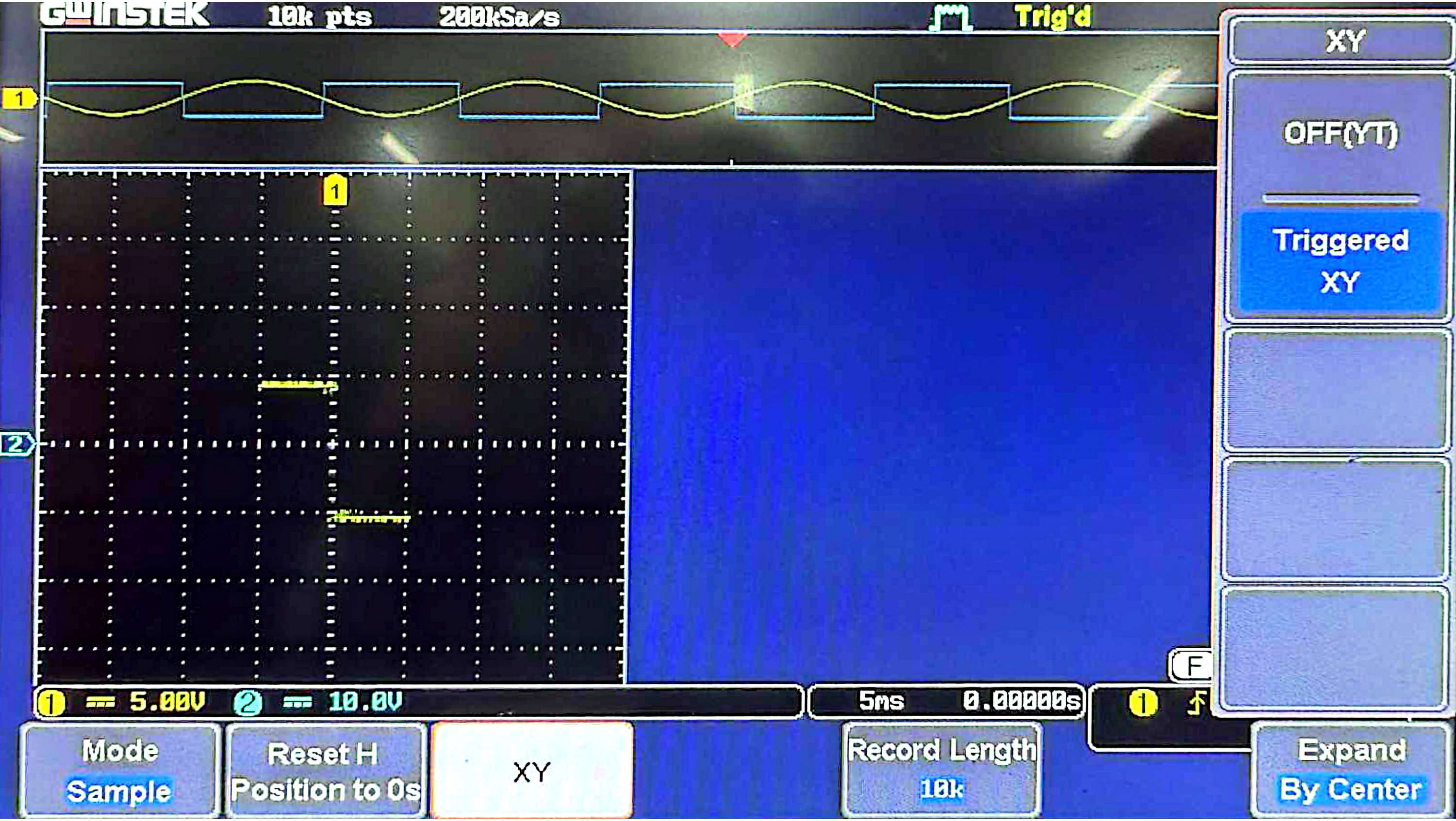


GW INSTEK
10k pts
200kSa/s
Trig'd
XY
OFF(YT)
Triggered XY
5.00V
10.0V
5ms
0.00000s
F
Mode
Sample
Reset H Position to 0s
XY
Record Length
10k
Expand
By Center

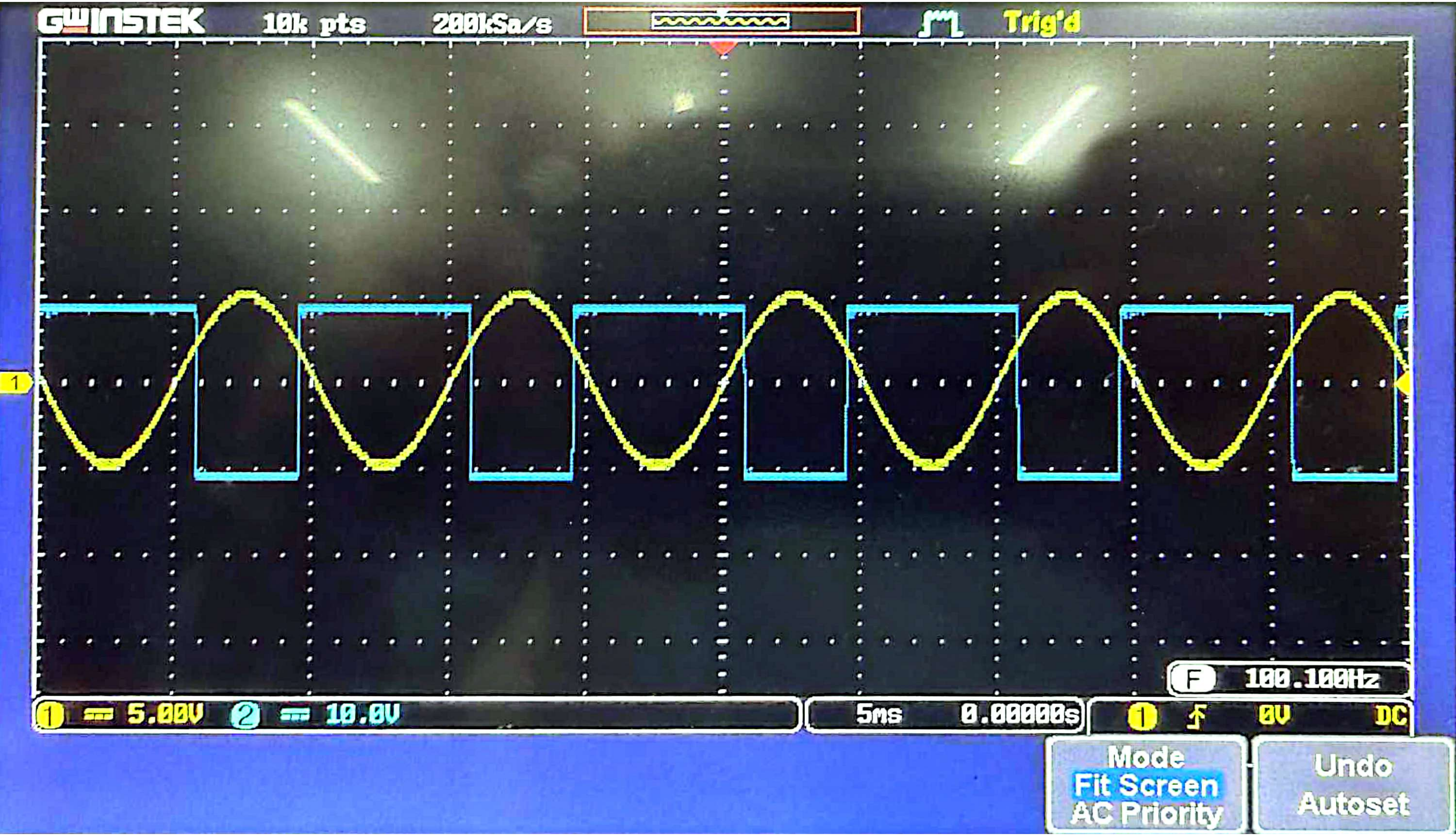

GWINSTEK
10k pts
200kSa/s
Tri
1
Measurement Summary
Pk-Pk 9.80V
Max 5.00V
Min -4.80V
Amplitude 9.60V
High 5.00V
Low -4.60V
Mean 172mV
CycleMean 170mV
RMS 3.47V
CycleRMS 3.47V
Area 8.62mVs
CycleArea 1.70mVs
ROVShoot 0.00%
FOVShoot 2.08%
RPREShoot 2.08%
FPREShoot 0.00%
Frequency 100.0Hz
Period 10.00ms
RiseTime 2.846ms
FallTime 2.987ms
+Width 5.050ms
-Width 4.950ms
Dutycycle 50.50%
+Pulses 4
-Pulses 5
+Edges 5
-Edges 4
1

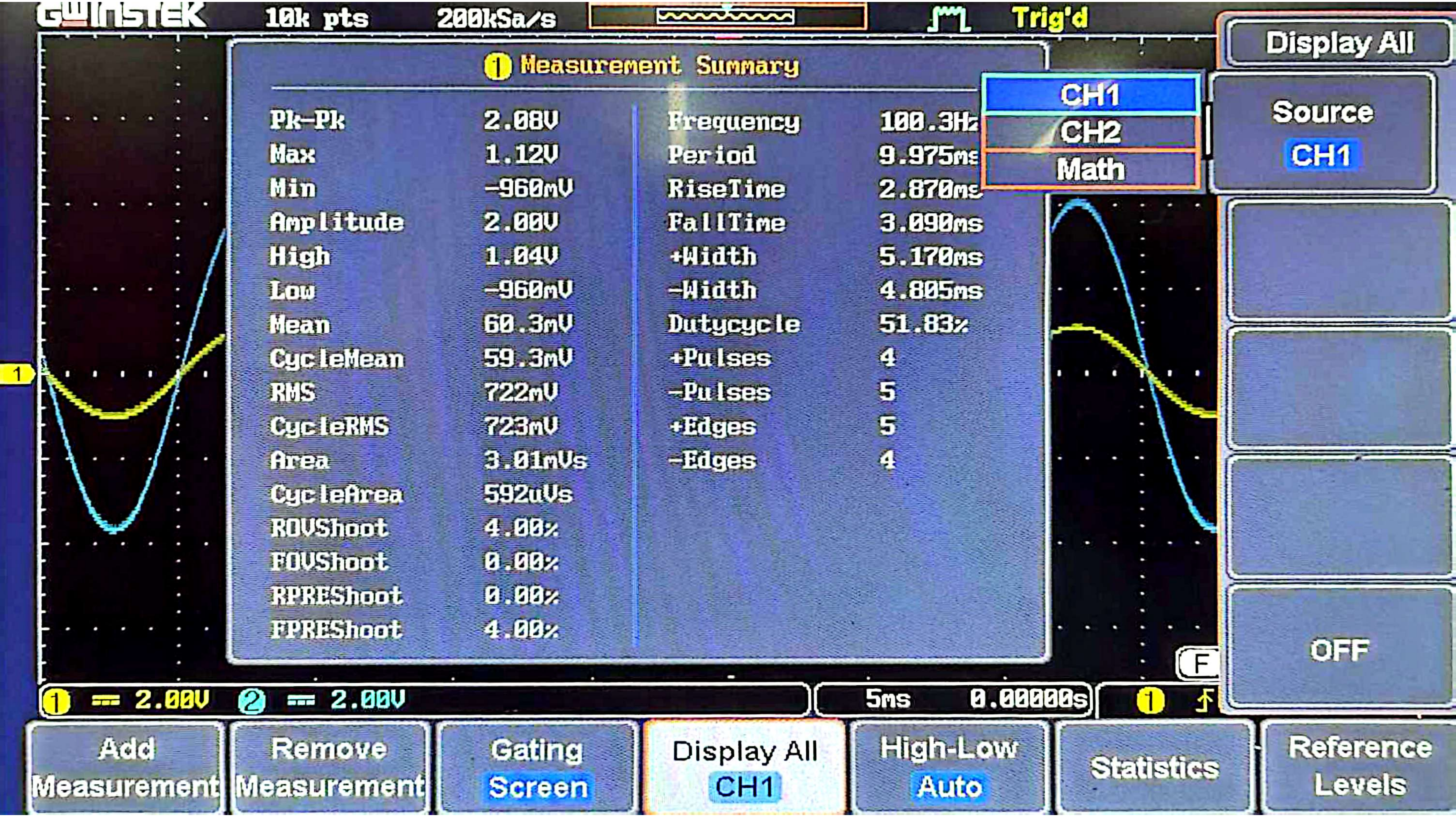

GW INSTEK
10k pts
200kSa/s
Trig'd
XY
OFF(YT)
Triggered XY
5.00V
10.0V
5ms
0.00000s
F
Mode
Sample
Reset H
Position to 0s
XY
Record Length
10k
Expand
By Center

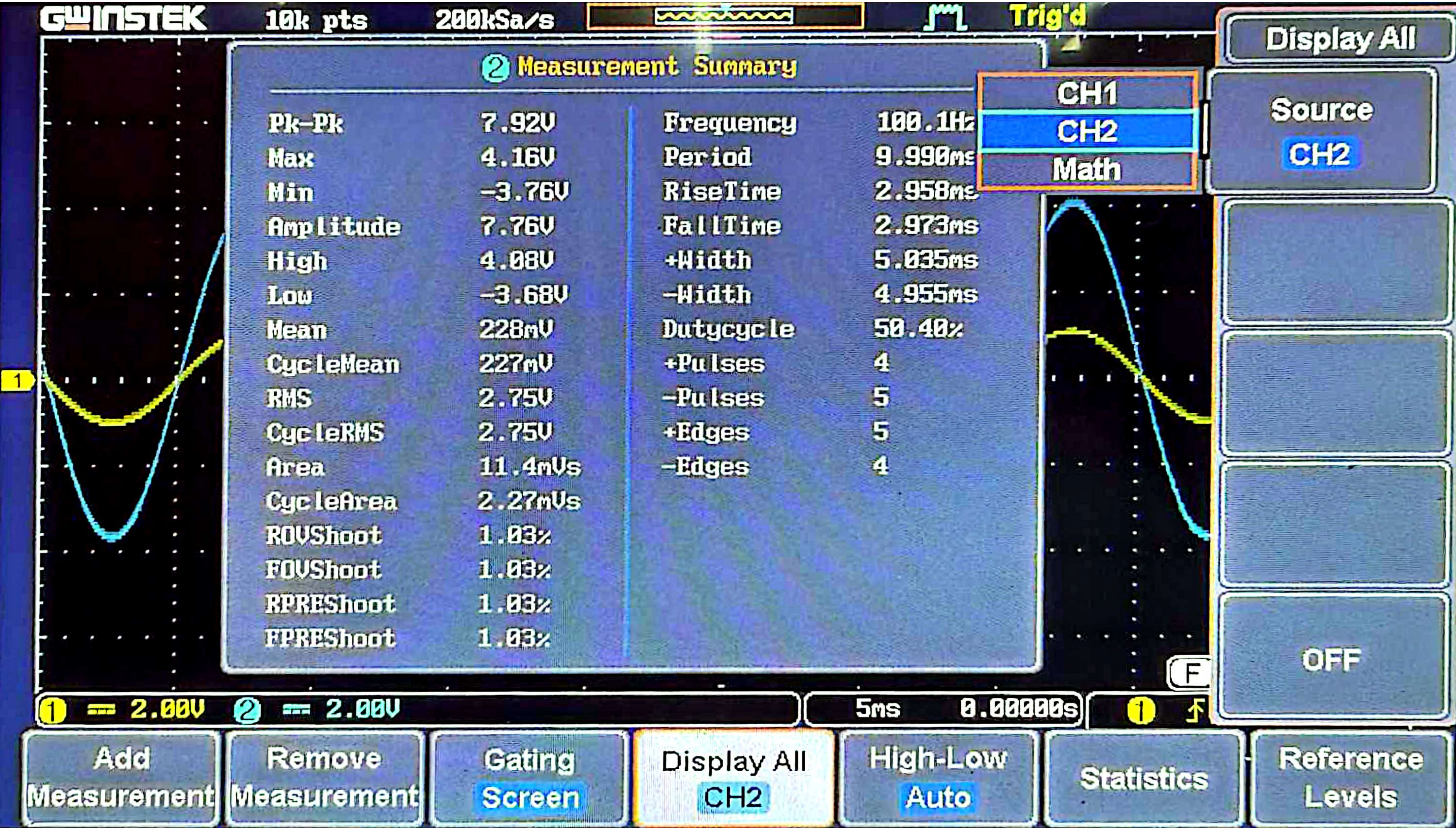

GW INSTEK
10k pts
200kSa/s
Trig'd
XY
OFF(YT)
Triggered XY
F
1 5.00V
2 10.0V
5ms
0.00000s
Mode
Sample
Reset H
Position to 0s
XY
Record Length
10k
Expand
By Center

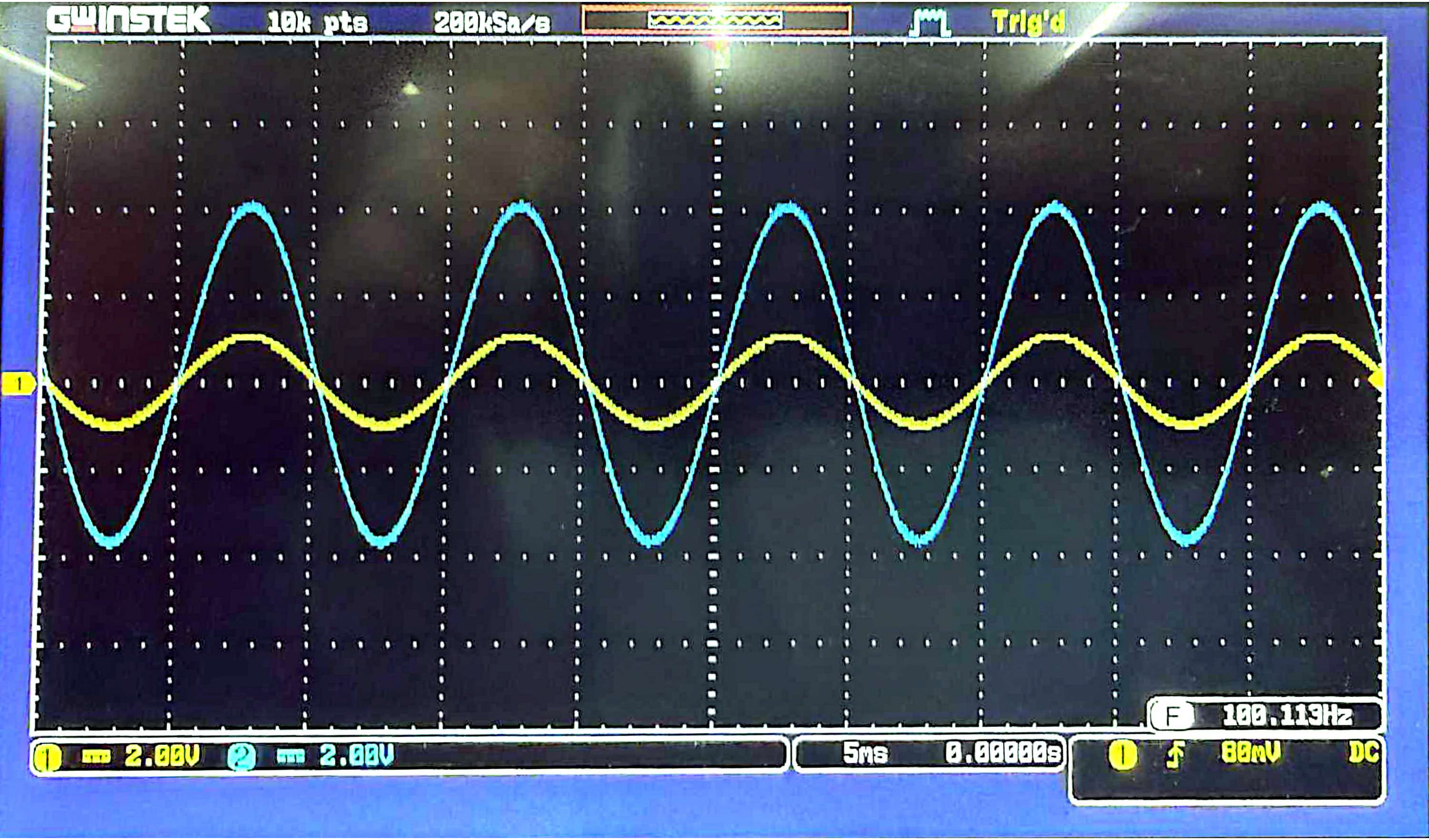


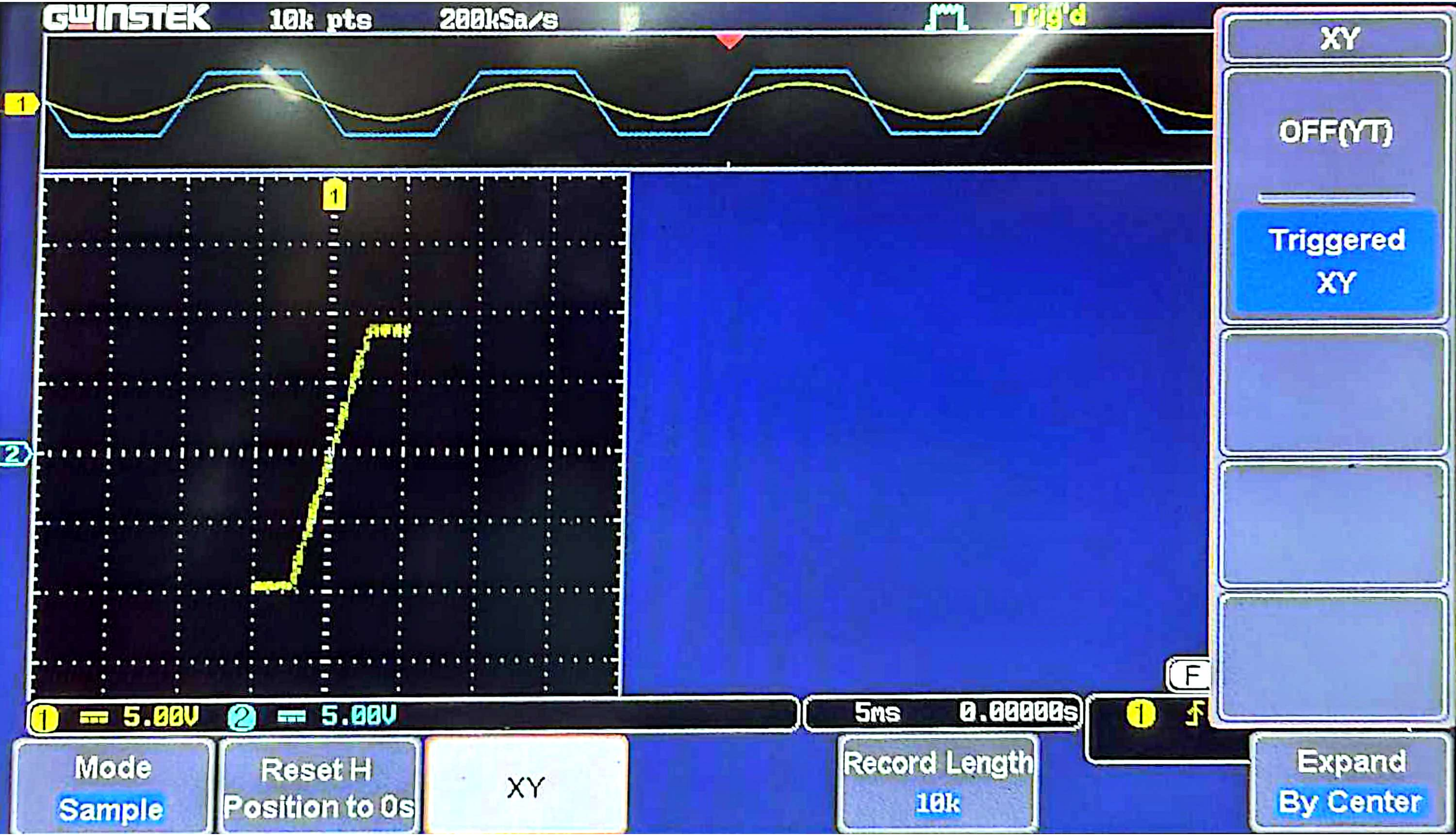
GW INSTEK
10k pts
200kSa/s
Trig'd
XY
OFF(YT)
Triggered XY
5.00V
10.0V
5ms
0.00000s
F
Mode
Sample
Reset H
Position to 0s
XY
Record Length
10k
Expand
By Center

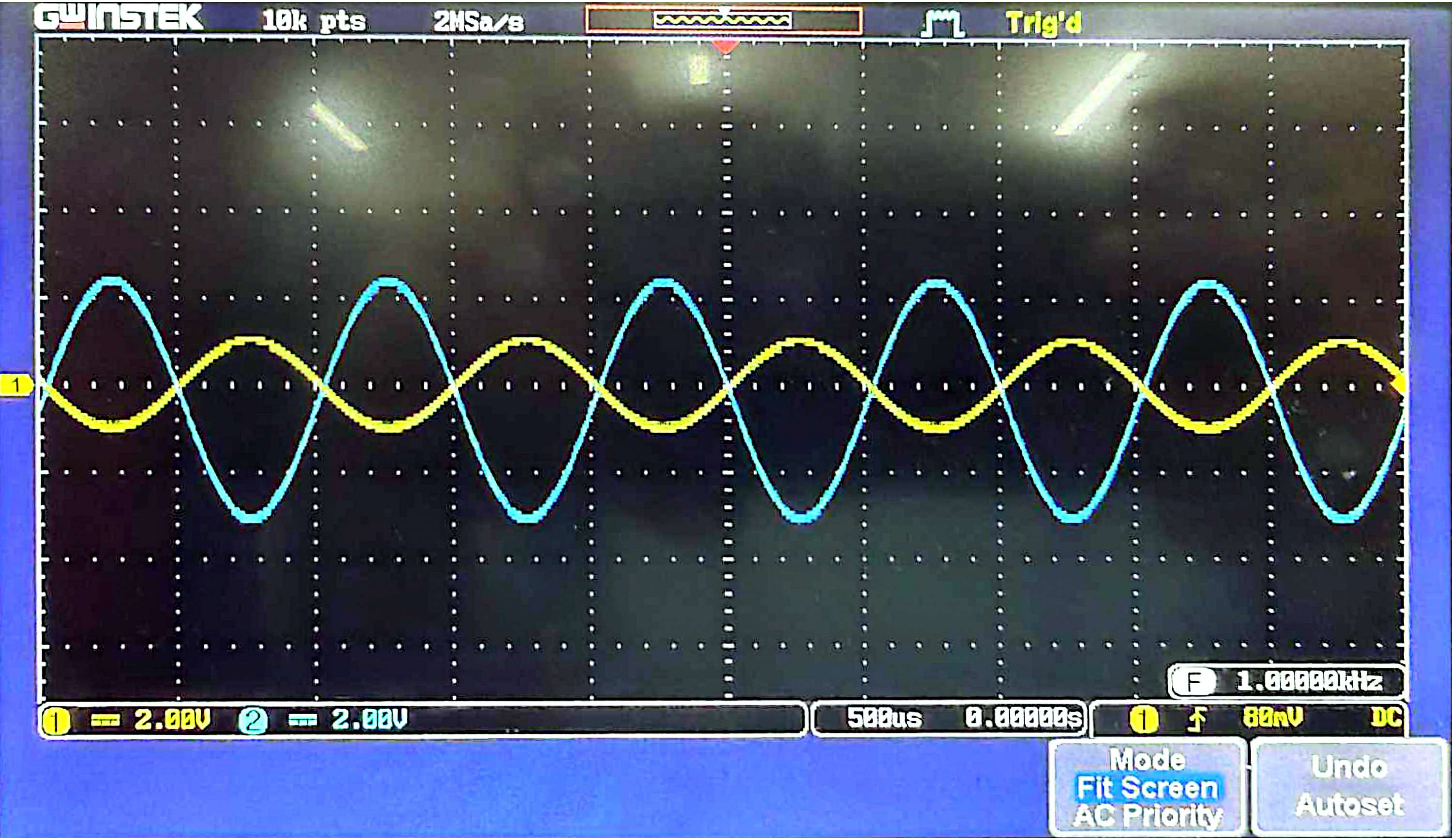

GW INSTEK
10k pts
200kSa/s
Trig'd
1
F 100.100Hz
1 5.00V
2 10.0V
5ms
0.00000s
1 0V DC
Mode
Fit Screen
AC Priority
Undo
Autoset

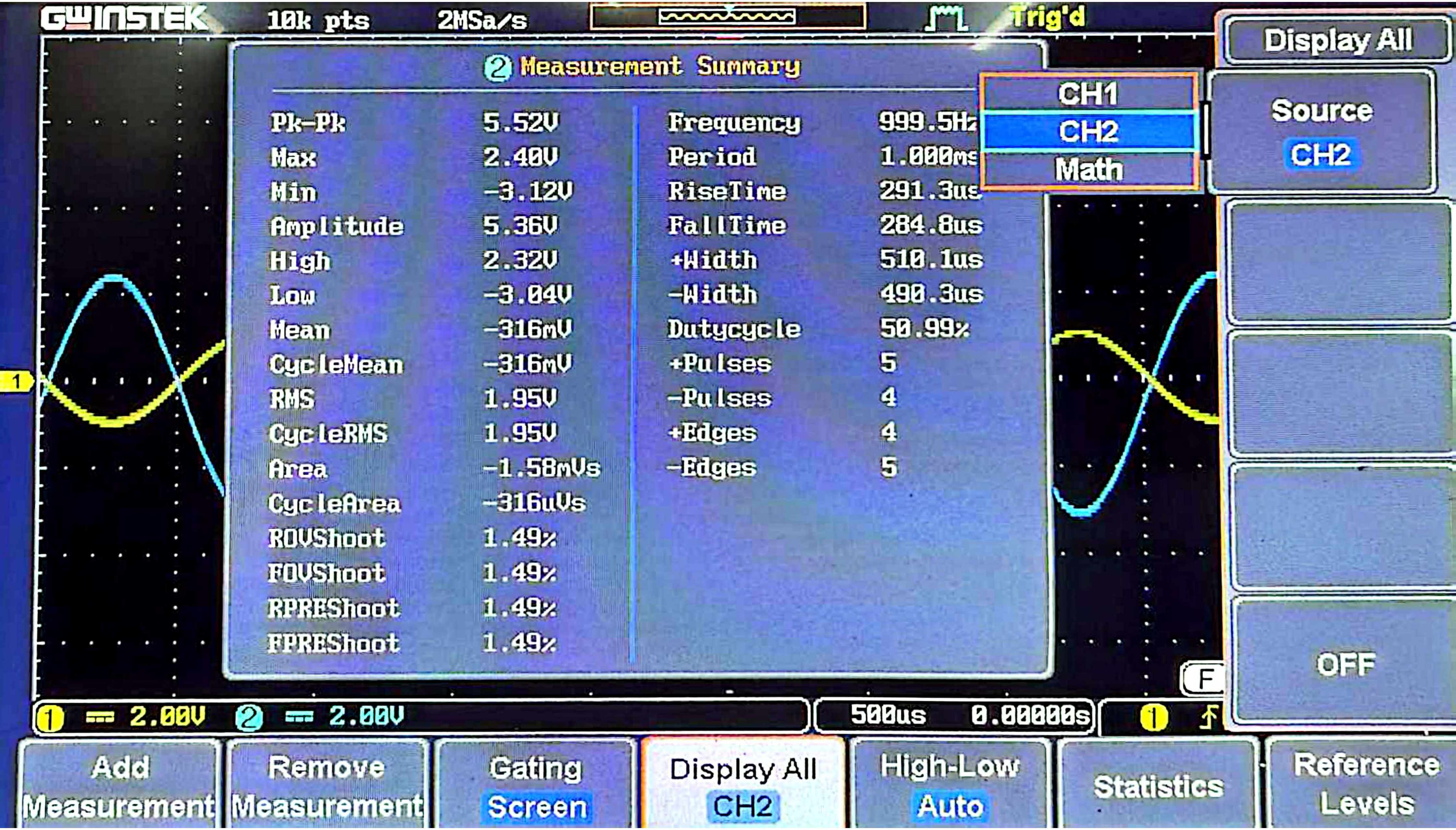

GW INSTEK
10k pts
200kSa/s
Trig'd
Measurement Summary
Pk-Pk 2.08V
Max 1.12V
Min -960mV
Amplitude 2.00V
High 1.04V
Low -960mV
Mean 60.3mV
CycleMean 59.3mV
RMS 722mV
CycleRMS 723mV
Area 3.01mVs
CycleArea 592uVs
ROVShoot 4.00%
FOVShoot 0.00%
RPREShoot 0.00%
FPREShoot 4.00%
Frequency 100.3Hz
Period 9.975ms
RiseTime 2.870ms
FallTime 3.090ms
+Width 5.170ms
-Width 4.805ms
Dutycycle 51.83%
+Pulses 4
-Pulses 5
+Edges 5
-Edges 4
CH1
CH2
Math
Display All
Source CH1
OFF
2.00V
2.00V
5ms 0.00000s
Add Measurement
Remove Measurement
Gating Screen
Display All CH1
High-Low Auto
Statistics
Reference Levels


GW INSTEK
10k pts
200kSa/s
Trig'd
Measurement Summary
Pk-Pk 7.92V
Max 4.16V
Min -3.76V
Amplitude 7.76V
High 4.08V
Low -3.68V
Mean 228mV
CycleMean 227mV
RMS 2.75V
CycleRMS 2.75V
Area 11.4mVs
CycleArea 2.27mVs
ROVShoot 1.03%
FOVShoot 1.03%
RPREShoot 1.03%
FPREShoot 1.03%
Frequency 100.1Hz
Period 9.990ms
RiseTime 2.958ms
FallTime 2.973ms
+Width 5.035ms
-Width 4.955ms
Dutycycle 50.40%
+Pulses 4
-Pulses 5
+Edges 5
-Edges 4
CH1
CH2
Math
Display All
Source
CH2
OFF
1 2.00V
2 2.00V
5ms
0.00000s
Add Measurement
Remove Measurement
Gating
Screen
Display All
CH2
High-Low
Auto
Statistics
Reference Levels

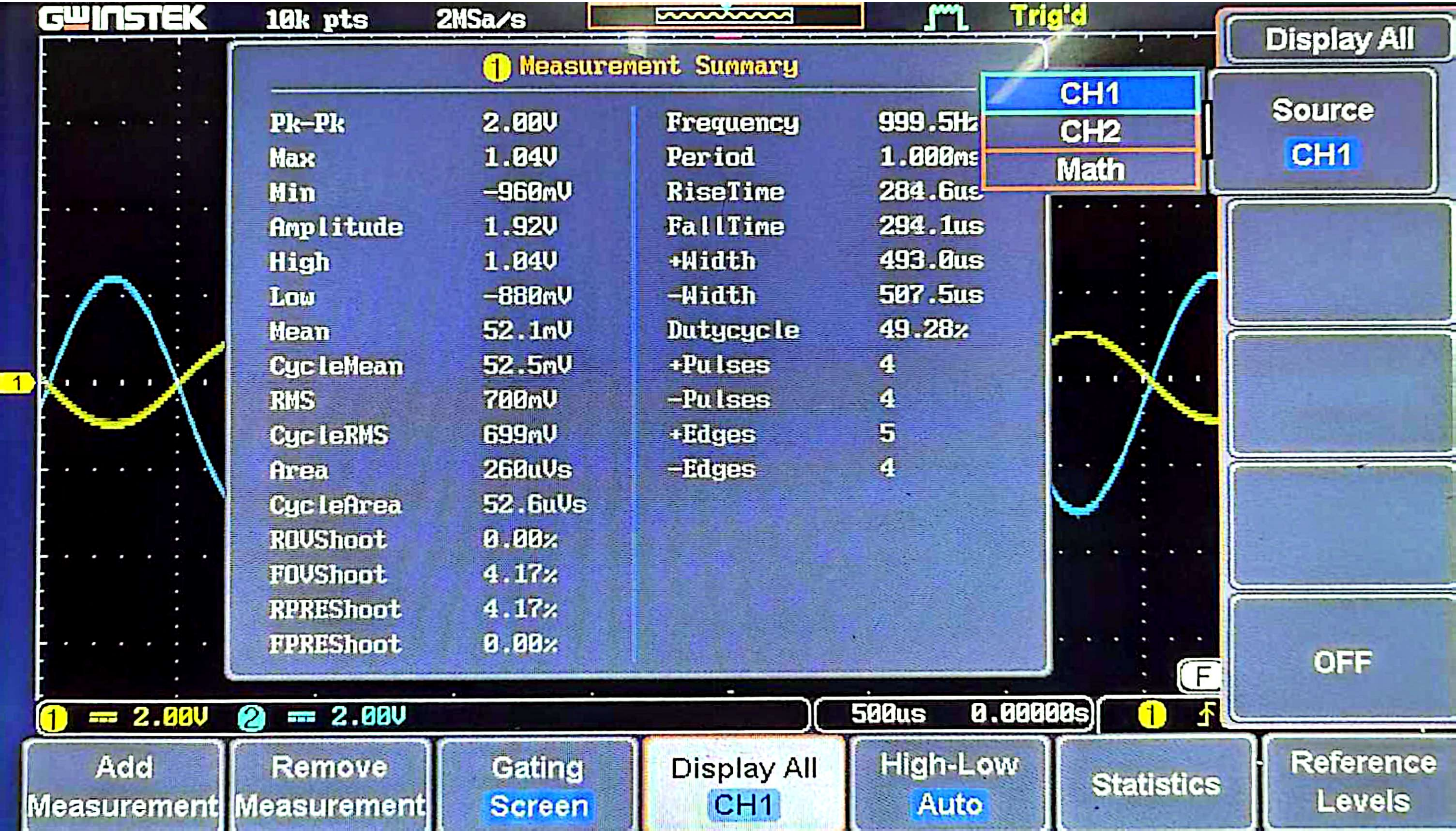

GWINSTEK
10k pts
200kSa/s
Trig'd
1
F 100.113Hz
1 2.00V
2 2.00V
5ms
0.00000s
1 80mV
DC

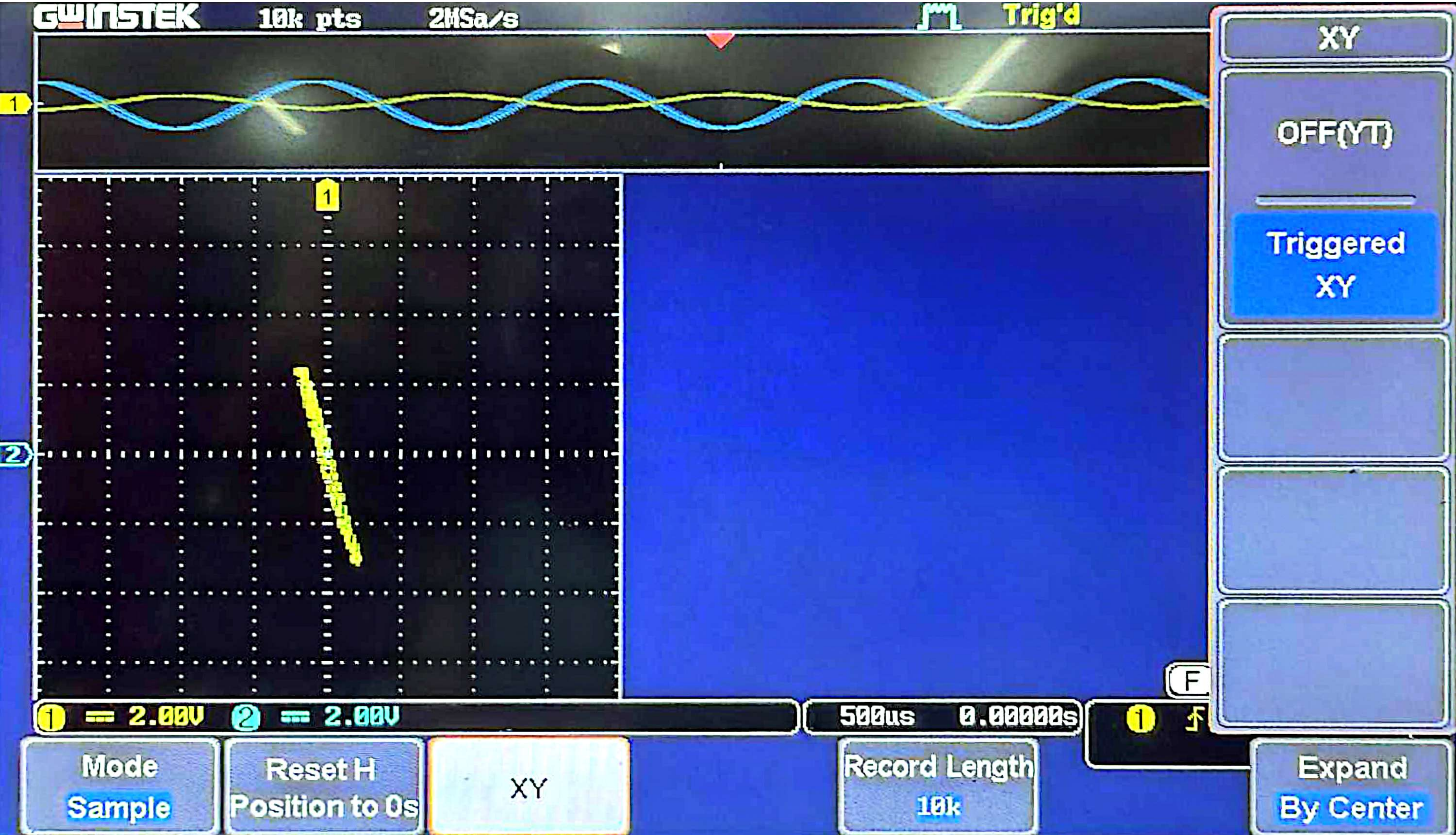

GWINSTEK
10k pts
200kSa/s
Trig'd
XY
OFF(YT)
Triggered
XY
5.00V
5.00V
5ms
0.00000s
Mode
Sample
Reset H
Position to 0s
XY
Record Length
10k
Expand
By Center


GW INSTEK
10k pts
2MSa/s
Trig'd
F 1.00000kHz
1 2.00V 2 2.00V
500us 0.00000s
1 80mV DC
Mode
Fit Screen
AC Priority
Undo
Autoset



GWINSTEK
10k pts
2MSa/s
Trig'd
Display All
2 Measurement Summary
Pk-Pk 5.52V
Max 2.40V
Min -3.12V
Amplitude 5.36V
High 2.32V
Low -3.04V
Mean -316mV
CycleMean -316mV
RMS 1.95V
CycleRMS 1.95V
Area -1.58mVs
CycleArea -316uVs
ROVShoot 1.49%
FOVShoot 1.49%
RPREShoot 1.49%
FPREShoot 1.49%
Frequency 999.5Hz
Period 1.000ms
RiseTime 291.3us
FallTime 284.8us
+Width 510.1us
-Width 490.3us
Dutycycle 50.99%
+Pulses 5
-Pulses 4
+Edges 4
-Edges 5
CH1
CH2
Math
Source
CH2
OFF
1 2.00V
2 2.00V
500us 0.00000s
Add Measurement
Remove Measurement
Gating Screen
Display All CH2
High-Low Auto
Statistics
Reference Levels


GWINSTEK
10k pts
2MSa/s
Trig'd
Measurement Summary
Pk-Pk 2.00V
Max 1.04V
Min -960mV
Amplitude 1.92V
High 1.04V
Low -880mV
Mean 52.1mV
CycleMean 52.5mV
RMS 700mV
CycleRMS 699mV
Area 260uVs
CycleArea 52.6uVs
ROVShoot 0.00%
FOVShoot 4.17%
RPREShoot 4.17%
FPREShoot 0.00%
Frequency 999.5Hz
Period 1.000ms
RiseTime 284.6us
FallTime 294.1us
+Width 493.0us
-Width 507.5us
Dutycycle 49.28%
+Pulses 4
-Pulses 4
+Edges 5
-Edges 4
CH1
CH2
Math
Display All
Source
CH1
OFF
1 2.00V
2 2.00V
500us 0.00000s
Add Measurement
Remove Measurement
Gating
Screen
Display All
CH1
High-Low
Auto
Statistics
Reference Levels



GWINSTEK
10k pts
2MSa/s
Trig'd
XY
OFF(YT)
Triggered XY
2.00V
2.00V
500us
0.00000s
Mode
Sample
Reset H Position to 0s
XY
Record Length
10k
Expand
By Center